

12.579

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Question

Let V be the vector space of all 3×3 matrices with complex entries over the real field. If

$$W_1 = \{ \mathbf{A} \in V : \mathbf{A} = \bar{\mathbf{A}}^T \}$$

$$W_2 = \{ \mathbf{A} \in V : \text{trace}(\mathbf{A}) = 0 \}$$

then the dimension of $W_1 + W_2$ is equal to _____.

The dimension of the sum of two subspaces is given by the formula:

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2) \quad (1)$$

The dimension of V , the space of 3×3 complex matrices over $\mathbb{R}$, is:

$$\dim(V) = 3 \times 3 \times 2 = 18 \quad (2)$$

Theoretical Solution

For W_1 (Hermitian matrices), the dimension is determined by 3 real diagonal elements and 3 complex upper-triangular elements.

$$\dim(W_1) = 3(1) + 3(2) = 9 \quad (3)$$

For W_2 (trace-zero matrices), the condition imposes two real constraints on the dimension of V .

$$\dim(W_2) = \dim(V) - 2 = 18 - 2 = 16 \quad (4)$$

Theoretical Solution

For $W_1 \cap W_2$ (Hermitian, trace-zero matrices), the trace condition imposes one real constraint on the dimension of W_1 .

$$\dim(W_1 \cap W_2) = \dim(W_1) - 1 = 9 - 1 = 8 \quad (5)$$

Substituting these values into the dimension sum formula:

$$\dim(W_1 + W_2) = 9 + 16 - 8 = 17 \quad (6)$$

Conclusion

The dimension of $W_1 + W_2$ is 17.